

DATA ANALYST: TASK 11

Topic: Skill Progression Analytics & Performance Benchmarking

Difficulty: Intermediate

Duration: 6 Hours

1. PROBLEM STATEMENT

Organizations need to measure how users progress across different skills and compare performance across colleges, departments, and learning categories. Data analysts must identify skill growth trends, benchmark performance, and provide recommendations to improve learning outcomes. The objective of this assessment is to analyze skill progression patterns, benchmark student performance, and create insightful reports that support data-driven decision making.

• 2. BUSINESS QUESTIONS TO ANSWER

Q1:

Which skill categories demonstrate the highest average scores?

Q2:

How do performance scores vary across departments?

Q3:

Which colleges show the strongest skill improvement trends?

Q4:

What percentage of students achieved Beginner, Intermediate, and Advanced proficiency levels?

Q5:

How does study time influence skill performance?

Q6:

Which skills have the highest failure rates?

Q7:

Which skill categories require additional learning interventions?

Q8:

What recommendations can improve skill development and performance?

• 3. TECHNICAL REQUIREMENTS

Dataset Preparation

Create or use a dataset containing:

- Student ID
- Student Name
- College Name
- Department
- Skill Category
- Skill Name
- Assessment Score
- Learning Hours
- Attempts Count

- Proficiency Level

Minimum Records

- 1000+ Students
 - 20+ Skills
 - 5+ Skill Categories
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Data Preparation

Perform:

Data Cleaning

- Missing Value Handling
 - Duplicate Removal
 - Standardization
 - Outlier Detection
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KPI Calculations

Calculate:

- Average Skill Score
 - Skill Completion Rate
 - Skill Failure Rate
 - Performance Growth %
 - Department Performance Index
 - College Benchmark Score
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Visualization Requirements

Create:

- Skill Distribution Chart
 - College-wise Performance Analysis
 - Department-wise Benchmark Dashboard
 - Skill Failure Analysis
 - Learning Hours vs Performance Chart
 - KPI Dashboard
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Tools Allowed

- Excel
 - Power BI
 - SQL
 - Python
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- **4. EXPECTED OUTPUT**

Performance Benchmark Dashboard

Display:

- ✓ Overall Average Score
 - ✓ Skill Completion Rate
 - ✓ Failure Rate
 - ✓ Performance Growth %
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Skill Analysis Report

Identify:

- ✓ Top Performing Skills
 - ✓ Weak Skill Categories
 - ✓ High Failure Areas
 - ✓ Department Comparison
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Benchmark Analysis

Compare:

- ✓ Colleges
 - ✓ Departments
 - ✓ Skill Categories
 - ✓ Proficiency Levels
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Visualizations

Generate:

- ✓ KPI Dashboard
 - ✓ Bar Charts
 - ✓ Pie Charts
 - ✓ Trend Charts
 - ✓ Skill Distribution Charts
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Recommendations

Provide:

- ✓ Skill Improvement Suggestions
 - ✓ High-Risk Skill Identification
 - ✓ Learning Strategy Recommendations
 - ✓ Performance Enhancement Plan
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- **5. SUBMISSION DELIVERABLES**

dataset.xlsx

Dataset used for analysis.

performance_analysis_report.pdf

Include:

- Skill Progression Analysis
 - Benchmark Metrics
 - Failure Analysis
 - Performance Insights
 - Business Recommendations
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visualizations/

Folder containing:

- Dashboard Screenshots
 - Charts
 - Skill Analysis Visuals
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INSIGHTS.md

Include:

- Executive Summary
- Key Findings
- Benchmark Analysis
- Recommendations

- **6. EVALUATION RUBRIC**

Assessment Criteria	Weight
Data Cleaning & Preparation	15%
KPI Calculation Accuracy	20%
Benchmark Analysis	25%
Visualization Quality	20%
Business Recommendations	20%

- **EXPECTED OUTPUT**

Excellent Standard

- ✓ Accurate KPI calculations.
 - ✓ Strong benchmark comparisons.
 - ✓ Professional dashboards and charts.
 - ✓ Meaningful business insights.
 - ✓ Actionable recommendations.
 - ✓ Well-structured documentation.
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Good Standard

- ✓ Correct analysis performed.
 - ✓ Basic benchmark reporting.
 - ✓ Proper visualizations.
 - ✓ General recommendations.
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- **LEARNING OUTCOME**

By completing this assessment, interns will gain practical experience in:

- Performance Benchmarking
- Skill Progression Analytics
- KPI Development
- Dashboard Design
- Comparative Analysis
- Business Intelligence Reporting
- Data-Driven Decision Making
- Stakeholder Reporting